

DINO-4DSTEM quick-start card

A one-page cheat sheet. See the full user manual for details.

Get a class map in 5 steps

1. **Data → Load + Pre-process:** Browse your file, **Load**. Set **vmax** so rings show but the centre is not a white blob.
2. Top bar: enter **real space** (nm/px) and **reciprocal** (nm⁻¹/px).
3. **Model → Training:** pick **Sample**, keep **Use paper recipe (locked)** *on*, click **Train**.
Or skip training: click the **run badge** (top bar) and load an existing model.
4. **Model → Eval:** **Render** to see the class map.
5. **Analysis → Post-hoc → Interpretation:** **Run interpretation** for figures + a written report.

Top bar (always visible)

- **dataset badge** : click to load your 4D-STEM file.
- **run badge** : click to load a trained model.
- **inference light** : on = a class map is ready.
- **real space / reciprocal** : your two calibration numbers (enter once; all plots scale automatically).
- **Inspect runs** : browse past models in a web page.
- **Assistant** : optional plain-English helper.

Which tab for what

- **Data** : open + prepare your file (vmax, crop, mask).
- **Model** : train, watch (Eval), apply to many files (Transfer).
- **Analysis** : explore the map (Post-hoc) and explain it (Interpretation).
- **Clustering** : classical comparisons (NMF, pretrained-DINO, SAM).
- **Diffraction** : peak finding (Blob) and orientation mapping (ACOM).

Common tasks

- **Map of my data with an existing model:** run badge → load model; dataset badge → load file; **Eval** → **Render**.
- **What do the classes mean:** **Analysis** → **Post-hoc** → **Interpretation** → **Run interpretation**.
- **See a class's clean diffraction:** **Post-hoc**, click a class on the map (shows the class average).
- **Compare to NMF:** **Clustering** → **NMF + cluster** → **Run**.
- **Crystal orientations:** **Diffraction** → **ACOM**: add a CIF, **Build all**, **Detect peaks**, **Single-phase full**, **Overlay**.
- **One model on a folder of files:** **Model** → **Transfer**, pick folder, tick files, **Run sequential transfer**.

Good habits

- Maximise the window before you start.
- A single low-dose pattern looks like noise; trust the **class average**.
- Changed crop/mask on **Data**? On **Training** click **Refresh crop/mask/COM from PrePanel** before **Train**.
- Buttons that “write new .cube.npy” make a new file; note its name.
- The “?” buttons next to controls explain each one in place.

If unsure of a setting for a new material, the locked “paper recipe” is a safe default. Don't close the small black log window: it keeps the program running.

Quick fixes

- **All white/black diffraction:** lower **vmax**, turn on **log-stretch**.
- **Empty / one-colour map:** make sure a model (run badge) and a class map (inference light) are loaded; click **Render**.
- **Nothing after Load:** check the black log window for a message.
- **Training feels stuck:** low-dose data loads slowly; give it time.